

1: Consider the following data set:

x	y
1	2.1
2	2.9
3	3.7
4	4.1
5	5.0

(i): Write a Matlab code using Least Squares Linear Regression to fit the linear model $y = a_0 + a_1x$.

(ii): Calculate the coefficient of determination (r^2).

(iii): Plot both the data and the fitted curve.

2: Consider the following data set:

x	y
0	1.0
1	2.7
2	5.8
3	10.1
4	17.2

(i): Write a Matlab code using Least Squares Linear Regression to fit the quadratic model $y = a_0 + a_1x + a_2x^2$.

(ii): Compare S_y and $S_{y/x}$.

(iii): Plot both the data and the fitted curve.
